

The Bright Side of Political Uncertainty: The Case of R&D

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We use close gubernatorial elections as a quasi-natural experiment to document a positive effect of political uncertainty on firm-level R&D. This finding is in contrast to the existing literature documenting a negative impact of political uncertainty on capital investment. We examine potential mechanisms and find that our results are consistent with the growth option view of R&D investment. The effect is stronger for politically sensitive and high-tech industries. The results are robust to different proxies for political uncertainty shocks. As predicted by models of investment under uncertainty, the real effects of political uncertainty critically depend on the type of the investment. (*JEL* G18, G38, O31, O32)

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A natural consequence of the political process is political uncertainty. Future government policy is largely unpredictable and changes over time with political turnover and shifting policy preferences. Recent research in finance and economics has provided evidence that political uncertainty has adverse effects on firm-level investment and harms economic performance.¹ These findings

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¹ For example, [Julio and Yook \(2012\)](#), [Baker, Bloom, and Davis \(2016\)](#), [Carvalho \(2018\)](#), [Gulen and Ion \(2016\)](#), and [Jens \(2017\)](#) show that real investment declines with higher uncertainty; [Colak, Durnev, and Qian \(2017\)](#)

are primarily motivated by the investment timing option in real options theory arguing that due to investment irreversibility, uncertainty exerts a strong negative effect on capital investment by increasing the value of waiting to invest (Bernanke 1983). The theoretical arguments and empirical findings have led researchers to conclude that policymakers should be mindful of the damage that political uncertainty may inflict on the economy (Gulen and Ion 2016).

While the empirical literature has provided ample evidence of a negative relationship between political uncertainty and irreversible investment, it has largely overlooked the effects of political uncertainty on other types of investment. It has also primarily focused on the option to delay, while the theory of investment under uncertainty suggests that other types of options, such as growth options, could exert a positive relationship between uncertainty and investment. We attempt to fill the gap in the existing literature by providing empirical evidence that political uncertainty can have a “bright side.”

Real options models such as Bernanke (1983), McDonald and Siegel (1986), and Stokey (2016) view investments as options and examine how uncertainty affects the value and timing of investment. In these models, because of non-convex-adjustment costs, uncertainty creates a valuable option to delay investment. A firm trades off the value of cash flows from early investment against the information value of waiting to invest because making mistakes under investment irreversibility is costly. Firms become inactive and adopt a “wait-and-see” approach when uncertainty increases, leading to the prediction of a negative relationship between uncertainty and investment. The importance of the expected value of bad outcomes under uncertainty in the value of waiting to invest is known as the “bad news principle” (Bernanke 1983). Consistent with these predictions, the empirical literature has found evidence that investments such as capital expenditures (Julio and Yook 2012; Jens 2017; Gulen and Ion 2016) are negatively related to political uncertainty.

A different strand of the theoretical literature argues, however, that some characteristics of investment, particularly those associated with research and development (R&D) projects, predict a positive relationship between uncertainty and investment. What we refer to as the growth options view, for example, considers R&D as multistage investments, where the firm makes expenditures to acquire an option to invest (or not) in the future.² The key distinction in these models that lead to a positive relationship between R&D and uncertainty is that an initial investment, such as research, is necessary for the firm to make subsequent investments.

find lower initial public offering (IPO) volume; Julio and Yook (2016) find declines in foreign direct investment; Gao, Murphy, and Qi (2019) find higher financing costs; and Chen, Jens, and Page (2023) and Bonaime, Gulen, and Ion (2018) document weaker merger and acquisition activity.

² In recent years, researchers have analyzed R&D investment in a contingent claims framework where R&D projects are viewed as investments that acquire options where the underlying asset is the cash flows to be received upon successful completion of the developed project (e.g., Berk, Green, and Naik 2004; Schwartz 2003; Childs and Triantis 1999).

Bar-Ilan and Strange (1996) show that if investments take time to develop with a lag between the initial investment and final investment, then uncertainty can have a positive effect on investment and argue that policymakers have less reason to be concerned about uncertainty. In their model, the cost of bad outcomes is bounded below because the firm can abandon the project in bad states of the world, but invest in good states. A firm that delays the initial investment cannot enter the market, so the opportunity cost of waiting depends on future outcomes rather than on the value of foregone cash flows in the case of the Bernanke (1983) model. Consequently, the opportunity cost of waiting increases with uncertainty, because the firm only invests in good states of the world and abandons its projects in bad states. The Bar-Ilan and Strange (1996) model predicts that a firm hit with a demand uncertainty shock may hasten investment rather than delay in the presence of these types of investment lags, suggesting a “good news principle” for investments such as R&D in contrast to the “bad news principle” in the Bernanke (1983) model.

Pindyck (1993) models investment as a compound option in which the firm is allowed to abandon the project midstream. In this model, R&D investment is characterized as having technical uncertainty, defined as uncertainty about the physical difficulty of completing the project, in addition to normal operating uncertainty. Pindyck (1993) shows that technical uncertainty makes investment more likely in the presence of an uncertainty shock to operations because investing reveals information about the project’s costs and therefore provides additional value to the firm. There is no information revelation without making the initial investment, so in the model there is no value to delaying investment, which leads to a positive association between R&D investment and uncertainty.

Since the investment/uncertainty relationship depends on the characteristics of the investment, the theory of investment under uncertainty suggests that within a given firm, some investments may be subject to delay in the face of a political uncertainty shock to firm operations, while others may be hastened in response to the same shock. For example, capital expenditures may be viewed as irreversible investments and more likely to be negatively related to uncertainty shocks because of the valuable option to delay. Investing in R&D, by contrast, gives a firm the option to invest if future conditions are favorable, or abandon if they are not. Under uncertainty, the growth option has a large potential upside and a limited downside, leading to a positive relationship between R&D and uncertainty. The purpose of this paper is to contribute to the existing literature by investigating the growth options hypothesis and test whether R&D investments respond differently from other types of investments to political uncertainty shocks.

Our empirical strategy employs different proxies for political uncertainty shocks. Our primary approach uses a quasi-natural experiment created by closely contested U.S. gubernatorial elections over the 1976 to 2020 period as a source of exogenous uncertainty shocks to a firm’s operating conditions.

Elections of state governors have implications for corporate decisions.³ State governors have substantial power in influencing state-level policymaking (e.g., [Peltzman 1987](#)). Close gubernatorial elections introduce political uncertainty about state leadership and almost all sorts of state government policies such as industry policy and regulation, fiscal policy, taxation, labor, customer and environmental protection that may directly or indirectly affect firms' R&D decisions.⁴

We focus on close election periods for two important reasons. First, firms consistently face political uncertainty and form expectations about the political environment in which they operate and close elections provide a shock to expectations about future policy changes and thus translate into uncertainty shocks about firm operations. Second, it is likely that political uncertainty is higher when an election is hotly contested and close election periods can be viewed as better proxies for political uncertainty ([Snowberg, Wolfers, and Zitzewitz 2007](#)). Following [Julio and Yook \(2012\)](#) and [Jens \(2017\)](#), we measure highly contested elections by using vote differences between the winner and the runner-up.

The timing of gubernatorial elections, unlike many elections in other countries around the world, is exogenously determined by law and not affected by general economic conditions. Using close gubernatorial elections as a source of political uncertainty, mitigates endogeneity concerns that changes in R&D investment may be caused by changes in business cycles or state economic conditions. Elections in different states occur in different years, which provides us with staggered cross-state and time-series variation in the timing of close gubernatorial elections to test our main hypothesis. Focusing our analyses on U.S. firms and close U.S gubernatorial elections, rather than on international elections, helps alleviate concerns that omitted variables could lead to a spurious correlation between R&D investment and political uncertainty, because U.S. firms located in different states share the same national political business cycles, have similar constraints in accessing capital markets, and share common cultural norms. A final advantage to using close elections to study the effects of political uncertainty on R&D investment is that other types of investments, such as capital expenditures, initial public offerings, and mergers and acquisitions have been studied using similar strategies, allowing for comparison of different effects.

In our empirical analysis, the treatment group consists of firm-years in the close-election year, while our control group is represented by firm-years after the election year, when uncertainty is mostly resolved, or before, when

³ A growing literature has documented that firms face political uncertainty before elections. See, for example, [Durnev \(2010\)](#), [Gao, Murphy, and Qi \(2019\)](#), [Julio and Yook \(2012, 2016\)](#), and [Jens \(2017\)](#).

⁴ According to the 2014 Global Innovation Index report, political uncertainty and government effectiveness under the political environment category are rated as the top-two most influential factors in affecting innovation activities. See <https://www.globalinnovationindex.org/userfiles/file/reportpdf/GII-2014-v5.pdf> for details.

the nature of the election is not yet realized. We find evidence consistent with the growth options view of R&D investment. We show empirically that political uncertainty shocks lead to an increase in firm-level R&D investment, controlling for firm characteristics, state macroeconomic conditions, time and firm fixed effects. The results are also economically significant. During close election periods, firms' R&D intensities, defined as the ratio of R&D expenditure to total assets, increase by an average of 2.62% (6.46%) over the sample mean (median) during election years, relative to the nonelection years. This finding, in contrast to the previous literature, implies that there is an economic bright side to political uncertainty.

The results are also robust to using quarterly, instead of annual data. The benefit of quarterly data is the higher frequency that allows us to examine more closely the dynamics of investment around the election date. The downside is that the sample period is shorter than the one for the annual data, and therefore it includes fewer election cycles. We again find a strong, positive effect of political uncertainty on R&D intensity in the three quarters leading up to a close election. We further compare variations in R&D expenditures to variations of capital expenditures and find robust evidence that the two types of investment behave differently in response to political uncertainty shocks. R&D expenditures increase, while capital expenditures decrease with the heightened political uncertainty associated with close election periods.

As an alternative to using close elections as proxies for uncertainty shocks and as an external validity test, we also employ an alternative measure of policy uncertainty. We use the state-level economic policy uncertainty (EPU-S) index of [Baker, Davis, and Levy \(2022\)](#) based on 3,500 local newspapers. We find similar results. R&D expenditures are positively correlated, while capital expenditures are negatively correlated with the EPU-S index over time.

Next, We turn to examining the potential mechanisms that drive our results. An important possible alternative to the growth options mechanism is the substitution mechanism. It is well documented in the literature that firms cut back on capital expenditures ([Julio and Yook 2012](#); [Jens 2017](#); [Gulen and Ion 2016](#)), decrease expenditures on mergers and acquisitions ([\(Nguyen and Phan 2017; Bonaime, Gulen, and Ion 2018\)](#) and increase cash holdings ([Jens and Page 2022](#)) when political uncertainty is high. Financing R&D is often more complicated than financing investment in physical capital and firms generally rely on internally generated funds or equity to finance R&D ([Brown, Fazzari, and Petersen 2009](#)). Therefore, it is plausible that the increase in R&D we observe in our results does not come from the option-like properties of R&D investment, but rather from firms using cash savings from reduced capital expenditures to finance R&D.

To test this hypothesis, we examine the degree of asset redeployability across firms, to see whether firms that hold back on capital expenditures are also the ones that increase R&D when uncertainty is high. [Kim and Kung \(2017\)](#) construct measures of asset redeployability and show that, consistent with the

theoretical predictions for irreversible investment, firms with less redeployable capital reduce investment more than firms with highly redeployable capital. If the substitution hypothesis represents the main mechanism driving our results, then we should observe that firms with less redeployable capital will increase R&D the most.

Using the [Kim and Kung \(2017\)](#) measures of redeployability, we do not find that the effect is concentrated among low redeployability firms. We also find that the effect of close elections on R&D investment lasts for three quarters, while the effect lasts only one quarter for capital expenditures. While we cannot completely rule out the substitution hypothesis, substitution between capital and R&D expenditures does not appear to be the main mechanism driving our results. Our results are consistent with the growth and abandonment option theories of R&D investment under uncertainty.

We provide several additional tests to further investigate the growth option hypothesis. Recognizing that not all types of R&D investment are the same, we examine the cross-sectional differences to identify firms that make R&D investments that are similar to those modeled in the literature. Specifically, we identify firms with R&D that look like growth options, have a high degree of technical uncertainty (i.e. the physical difficulty of completing an R&D project), and have projects that are likely to be completed in stages. R&D projects with a high degree of technical uncertainty and long time-to-build may create valuable options on investment that outweigh uncertainty's discouraging effects due to irreversibility and in turn lead to early investments (e.g., [Grossman and Shapiro 1986](#); [Pindyck 1993](#); [Bar-Ilan and Strange 1996](#)). Building on this intuition, we expect that the growth option effect is larger for firms operating in industries, in which R&D projects' technical uncertainty tends to be high and the product development cycle is usually long. We also examine firms whose value comes largely from future growth options. Consistent with the theoretical literature, we find that the positive relation between political uncertainty about future government policies and R&D intensity is stronger for these subsamples of firms. We also find that the effect of political uncertainty on R&D is stronger for firms in more politically sensitive industries.

Our paper contributes to a growing body of literature on the role of politics, especially political uncertainty, in shaping firm performance and corporate decisions. To the best of our knowledge, we are the first to relate firm-level R&D dynamics to political uncertainty created by close U.S. gubernatorial elections. Despite the relatively large theoretical literature, there is very little empirical research on the relationship between uncertainty in general and R&D expenditures. While recent research largely suggests that political uncertainty about government policies has negative real effects, we are one of the first to document a bright side of political uncertainty. A notable exception is [Stein and Stone \(2013\)](#), who use an instrumental variables approach to identify the effect of general economic uncertainty on firm hiring, investment, advertising,

and R&D expenditures. They show that while investment and hiring declines when uncertainty is high, R&D moves in the opposite direction. Our results complement those in [Stein and Stone \(2013\)](#), showing that political uncertainty also appears to encourage R&D.

1. Data

1.1 Gubernatorial elections

Our primary measure of political uncertainty shocks is close gubernatorial elections in the United States. The primary source of the U.S. gubernatorial election data is the Congressional Quarterly (CQ) Press Electronic Library. The timing of U.S. gubernatorial elections is fixed by law. Every state but Louisiana holds its gubernatorial election on the first Tuesday following the first Monday in November.⁵ The vast majority of the states hold gubernatorial elections every 4 years, with the exception of Vermont and New Hampshire, which choose to run their gubernatorial elections every 2 years. Five states, including Louisiana, Kentucky, Mississippi, New Jersey, and Virginia, elect their state governors in odd-numbered years just preceding a presidential election. Other states run their gubernatorial elections in even-numbered years to coincide either mid-term elections or presidential elections. In 38 states, governors are limited to two consecutive terms. In some cases, states have changed the length of their gubernatorial election cycle. For example, the state of Arkansas and the state of Rhode Island switched from a 2-year election cycle to a 4-year election cycle in 1986 and 1994, respectively.⁶

An important feature of gubernatorial elections, unlike presidential elections, is that they occur in different states and different years. Therefore, substantial across- and within-state variations exist in addition to the time-series variation in the timing of gubernatorial elections. There are total 598 gubernatorial elections during the sample period of 1976 to 2020.⁷ Furthermore, there are only 10 presidential elections during the same period, which is not an adequate sample to yield any meaningful statistical inferences.

Our data contain 598 gubernatorial elections held between 1976 and 2020. Detailed election information is obtained from a variety of sources. The primary source for election and regime change data is the CQ Press Voting and Elections Collection, which is part of the CQ Press Electronic Library.⁸

⁵ The election timing of Louisiana may differ every year due to the adoption of the open primary system, where all the candidates for an office run together in one election. See Wikipedia for more detailed discussion about elections in Louisiana: http://en.wikipedia.org/wiki/Elections_in_Louisiana.

⁶ The only special election in the sample period took place in California in 2003. It resulted in voters replacing incumbent Democratic Governor Gray Davis with Republican Arnold Schwarzenegger. We treat this observation as all other election observations, and its inclusion has no effect on the results.

⁷ In unreported analysis, we exclude New Hampshire and Vermont, which follow a 2-year gubernatorial term, and find that the results are not affected.

⁸ The CQ Press Electronic Library database is available at <http://library.cqpress.com/elections/>.

Table 1
Summary statistics

A. Election characteristics

	N	Mean	Median	SD
Gubernatorial elections	598			
Incumbent Republican	287			
Incumbent Democrat	303			
Incumbent other	8			
Victory margin (%)	598	16.13	12.67	13.52
Close election (%)	138	2.59	2.65	1.43
Incumbent reelected	281			

B. Annual sample firm characteristics (close election sample)

	N	Mean	Median	SD
R&D intensity	26,262	0.0876	0.0356	0.1503
<i>q</i>	26,262	2.2201	1.5143	2.4729
Cash flow	26,262	-0.0389	0.0652	0.3727

C. Quarterly sample firm characteristics

	N	Mean	Median	SD
R&D intensity	112,333	0.0121	0.0000	0.0265
Capital expenditures/assets	112,333	0.0157	0.0088	0.0209
<i>q</i>	112,333	2.2623	1.5279	3.3986
Cash flow	112,333	0.0017	0.0188	0.0957

D. Average annualized realized stock return volatility

Nonelection quarter	0.4192			
Election quarter	0.4614			
Nonclose elections		0.4552		
Close elections		0.4687		
<i>t</i> -test of difference	33.995	4.972		

Panel A reports summary statistics for gubernatorial elections held between 1975 and 2020. Panel B reports summary statistics for the annual firm-level data in the analysis. Panel C reports summary statistics the quarterly firm-level data. Panel D reports the annual mean R&D intensity during nonelection quarters, election quarters, and close-election quarters. See the [appendix](#) for variable descriptions as well as the variable sources.

This database contains information on election date, the names of republican, democrat and independent candidates (if any), incumbent party affiliation, whether the incumbent governors seek reelection, whether the incumbents are subject to term limits, other reasons for the incumbents not to participate in the election (e.g., defeated in primary or retired or simply not running for reelection), the winning candidate party affiliation, the percentage vote for each candidate and the vote margin of the election. We supplement the election data with hand-collected information from Wikipedia.

Panel A of Table 1 summarizes our gubernatorial election data for the sample period from 1976 to 2020. There are 598 gubernatorial elections in total, distributed fairly evenly across the 50 U.S. states. Following [Julio and Yook \(2012\)](#) and [Jens \(2017\)](#), we classify an election as a close election if the victory margin, defined as the percentage vote difference between the first and second place candidates, is less than or equal to 5%. As a second measure of close elections, we consider if the incumbents are reelected. We expect elections where incumbents are not reelected to be more uncertain. Of the 598 elections,

138 are defined as close, using the margin of victory measure. The average close election has a vote difference of 2.59%, compared to the sample average voting difference of 16.13%. In 317 elections, incumbent governors are not reelected.

In this paper we focus on close election cycles. A close-election cycle is a 4-year period around a close election $[-1, 0, +1, +2]$, where the close election happens at the end of year 0. Close election years represent a significant shock to political uncertainty. Therefore, observations in the year of the close election comprise our treatment group and observations in the years before and after, represent the control group in our differences-in-differences analysis in the empirical section.

1.2 Firm- and state-level data

We obtain firm characteristics data from Compustat North America Fundamentals Annual files for the period from 1976 to 2021. Our sample runs from 1976 as the accounting treatment of R&D expense reporting was not standardized by Financial Accounting Standards Board until 1975.⁹ We also conduct analysis on Compustat quarterly data. The annual data allow for a longer time horizon and more elections, but the quarterly data allow for a higher frequency investigation of the dynamics of investment around the election date. The quarterly data span the period from 1988 to 2021.

R&D expenses represent all costs incurred during the financial year that relate to the development of new products or services. In this paper, the main variable of interest is R&D intensity, measured as the ratio of firm's R&D expense to its total assets. To isolate the effect of political uncertainty and firm R&D policies, we control for the standard set of control variables, following [Atanasov \(2013\)](#) and [Kogan et al. \(2017\)](#). Panels B and C of Table 1 report summary statistics for the annual and quarterly Compustat samples, respectively.

Our sample contains all domestic firms listed on NYSE, AMEX, and NASDAQ that are in close-election periods. Since we analyze the relation between political uncertainty and firm R&D investment, we also include in the analysis all the over-the-counter (OTC) traded domestic firms, which tend to be small technology stocks.¹⁰ Our main results focus on annual R&D expenditures, since quarterly R&D data are generally unavailable until 1989. We repeat the analysis on quarterly data and find consistent results. The sample includes firm-year combinations with nonmissing R&D expense and control variables.¹¹ We also require total assets to be no less than \$5 million. Following [Aghion et al. \(2005\)](#), we control for industry concentration using the *Herfindahl*

⁹ Financial Accounting Standards Board Statement No. 2.

¹⁰ In the robustness tests, we show that the exclusion of OTC-traded firms does not alter our results.

¹¹ In unreported robustness checks, we set missing R&D expense to zero and find broadly similar results.

index calculated over three-digit SIC code industries. We use the squared Herfindahl index, *Herfindahl*¹², to control for possible nonlinear (inverted-U-shaped) effects of industry concentration. For the annual data set, we end up with a sample of 26,262 firm-year observations between 1976 and 2021, during close election periods. Panel B of Table 1 reports summary statistics for the firm-level characteristics used in the annual empirical analysis. For the quarterly data set, we have a sample of 112,333 firm-quarter observations between 1988 and 2021, during close election periods. Panel C of Table 1 reports summary statistics for the firm-level characteristics used in the quarterly empirical analysis.

We combine accounting data from the Compustat database and the U.S. gubernatorial election data by election year and firm headquarter state.¹² There is a potential issue with this approach in that Compustat only reports the current state of a firm's headquarters, not its historical headquarters, which introduces measurement error if the firm has relocated. However, the number of firms that relocate is on average small and should introduce only a small amount of noise to our results.¹³

In unreported tests, we also show that our results are robust to using an alternative measure of firm's headquarter state location based on García and Norli (2012). They go through 10-K statements and count how many times a given state is mentioned, excluding states that are mentioned for other reasons such as Delaware. Arguably, the number of times a state is mentioned strongly correlates with the state-level operations of individual firms, since the headquarter state is not necessarily the state where the firm conducts the majority of its business. The García and Norli (2012) time-varying data allow us to track when firms change their state of operations. The data also permit assigning more than one state to a firm as many firms have large-scale operations in several states and could be affected by the uncertainty in each of those states.

State-level macroeconomic conditions are included in some of the main regressions to account for the general economic conditions within a state. We obtain the annual change in state GDP and annual state unemployment rate from the U.S. Bureau of Economic Analysis (BEA) website.¹⁴ All continuous variables are winsorized at the 1st and 99th percentiles to remove the influence of extreme outliers. Table A1 in the appendix defines the variable descriptions as well as the variable sources in detail.

¹² Howells (1990) and Breschi (2008) show that firms usually locate their R&D facilities close to headquarters and do not disperse them geographically.

¹³ For example, Pirinsky and Wang (2006) find only 118 examples of relocation in a sample of more than 5,000 firms over 15 years. Heider and Ljungqvist (2015) show that only 4% of firms' headquarter states are recorded incorrectly in Compustat for the fiscal year 2011.

¹⁴ BEA website is available at <http://www.bea.gov/>.

1.3 Return volatility around elections

To validate the use of elections as proxy for variation in political uncertainty, we use stock return data from the Center for Research in Security Prices (CRSP) to measure annualized return volatility for the firms in the CRSP/Compustat intersection. We use daily return data to measure volatility as the annualized standard deviation of stock returns. Panel D of Table 1 reports summary statistics for return volatility for different election periods and industries. The first column shows that return volatility is 0.4192 in nonelection quarters and 0.4614 in election quarters. The difference is statistically significant. The second column of panel D demonstrates that that average return volatility is higher during close elections.

2. Empirical Results

2.1 R&D investment around close-election periods

We investigate corporate R&D policy in a multivariate panel setting to control for firm characteristics and economy-wide events. We employ a difference-in-differences (DD) framework to evaluate changes in corporate R&D intensity when a firm experiences a political uncertainty shock in the form of a highly contested gubernatorial election.

The degree of political uncertainty depends to a large degree on the predictability of the election outcome. Highly unpredictable elections introduce exogenous shocks to uncertainty about a firm's operating environment and are considered better proxies for political uncertainty (Snowberg, Wolfers, and Zitzewitz 2007). Consistent with the idea that political uncertainty is higher during close elections, Kelly, Pástor, and Veronesi (2016) show that election-related uncertainty is reflected in the price of options. They find that implied volatility, the slope of implied volatility, and the variance risk premium, all measures of the value of protection against political uncertainty, increases around elections and that the effects are significantly larger for close elections. Jens (2017) find that contractions in capital investment during gubernatorial election periods are larger when the elections are close. We test whether increased uncertainty as proxied for by close elections is reflected in firm R&D expenditures.

Our main measure of close elections is based on the margin of victory, defined as the difference in the realized vote percentages between the first and second place candidates. An election is considered to be close if the margin of victory is less than 5%. Of the 598 elections covered in our analysis, 138 (approximately 23%) are identified as close elections. In the next section, we also compare the results of close-elections with the results of nonclose elections. We also show that our close-election ex post measure is positively correlated to ex ante measures based on polling data.

To test our hypothesis for the impact of close elections on R&D investment, we focus on a sample of firm-years that represent close election cycles.

We classify each 4-year election cycle $[-1, 0, +1, +2]$ as close/nonclose based on the margin of victory in the election year so that the split samples are mutually exclusive and collectively exhaustive. We compare R&D investment by firms in close election (high political uncertainty) years (year 0) to the rest of their R&D investment in nonelection years, when the uncertainty is lower.

The main regression model is specified as follows:

$$\text{R\&D Intensity}_{ijt} = \alpha_i + \gamma_t + \beta_0 \times \text{Close election dummy}_{jt} + \sum \varphi_i \mathbf{X}_{it} + \epsilon_{ijt}, \quad (1)$$

where i indexes firms, j indexes states, and t indexes years. The dependent variable is the firm-level R&D intensity, measured as the ratio of R&D expenditures to total assets. The main variable of interest is the close election dummy, $\text{Close election dummy}_{jt}$, which takes on a value of one if a gubernatorial election occurred in state j in year t , during a close-election cycle, where the margin of victory is less or equal to 5%, and zero if there is no election in state j in year t , during a close-election period. The above DD model uses firms in states in years without an election as the control group for a treated sample of firms in states in election years. The coefficient estimate on the election dummy, β_0 , is thus designed to capture changes in R&D intensity between election and nonelection years.

The intuition behind our empirical strategy is as follows. We consider an election year in a close election cycle as a significant shock to political uncertainty. In that case, firm-years that are subject to the close-election shock represent the treatment group, for which the political uncertainty is increased. Firm-years after the election years, when the uncertainty is resolved, and firm-year before the election year, before it is realized that it is a close election, comprise the control group, for which political uncertainty is lower.

To control for firm characteristics, we include a set of control variables motivated by [Aghion et al. \(2005\)](#) and [Atanassov \(2013\)](#), who identify potential determinants of R&D investment, in both the cross-section and over time. $\mathbf{X}_{it}$ is a vector of firm characteristics, which include Tobin's q , Cash flow, $\ln(\text{Age})$, $\ln(\text{Sales})$, Tangibility, Profitability, and Leverage. In some tests, for robustness, we also control for some state-level macroeconomic variables, including annual state GDP, GDP growth rate and the state unemployment rate. In addition, we include both firm and year fixed effects in the baseline R&D regression to account for any time-invariant unobservable factors. The specification captures within-firm variation in corporate R&D intensity around close gubernatorial election event years. We compute heteroscedasticity-adjusted standard errors clustered by state in the first four specifications following [Bertrand, Duflo, and Mullainathan \(2004\)](#), who argue that the clustering should be done by the unit of observation of the main explanatory variable. In our case, our main explanatory variable is the election indicator that varies by state. For robustness, we adjust the standard errors for by firm-state cluster in column 5 and by state and year cluster in column 6, following [Petersen \(2009\)](#).

Table 2
Political uncertainty and R&D intensity: Baseline results

	Dependent variable: R&D intensity					
	(1)	(2)	(3)	(4)	(5)	(6)
Close election dummy	0.0025*** (0.0009)	0.0025*** (0.0009)	0.0025*** (0.0009)	0.0023*** (0.0009)	0.0023*** (0.0009)	0.0023** (0.0011)
q	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0008)	0.0040*** (0.0011)
Cash flow	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416*** (0.0148)	-0.0416** (0.0163)
ln(Sales)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)	0.0014 (0.0016)
ln(Age)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0043)	0.0101** (0.0045)
Tangibility	0.1054*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0147)	0.1053*** (0.0149)
Profitability	-0.2244*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0114)	-0.2245*** (0.0151)
Leverage	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0127)	-0.0262** (0.0129)
Herfindahl		-0.0039 (0.0025)	-0.0039 (0.0025)	-0.0038 (0.0026)	-0.0038 (0.0026)	-0.0038 (0.0039)
GDP growth				0.0425 (0.0276)	0.0425 (0.0276)	0.0425 (0.0340)
Unemployment				-0.0001 (0.0008)	-0.0001 (0.0008)	-0.0001 (0.0008)
N	25,177	25,177	25,177	25,177	25,177	25,177
R ²	.8447	.8447	.8447	.8447	.8447	.8447
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The main explanatory variable is the close election year dummy (year 0). The sample is over close election periods (margin of victory is less than 5%). Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in parentheses below coefficient estimates, are clustered at the state level following [Bertrand, Duflo, and Mullainathan \(2004\)](#) in columns 1–4, at the state and firm levels in column 5, and at the state and year levels in column 6, and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 2 summarizes the results for our baseline R&D regression specification. In columns 1 to 6, we sequentially include additional variables describing firm characteristics and state macroeconomic conditions. The coefficient estimates on the close election year dummy are positive and statistically significant across all specifications, consistent with the theory predicting that political uncertainty stimulates firm R&D investment. Depending on the specification, the increases in conditional R&D intensity range between 0.0023 and 0.0025. The estimates reported in column 4, which represent the baseline specification throughout the rest of the analysis, show that R&D intensities increase by 0.0023 on average in close election cycles, after controlling for firm and state variables. The signs on the estimated coefficients of the control variables are consistent with previous findings in the literature. For brevity, in subsequent analyses we do not report coefficient estimates for the control variables.

In terms of economic magnitude, the coefficient estimate on the election dummy in column 4 of Table 2 translates into an economically significant

2.63% (6.46%) increase¹⁵ in firms' R&D-to-assets ratio in election years, relative to the mean (median) R&D intensity in nonelection years during close-election periods. We note that the magnitude of the increase in R&D investment in the close election year is slightly smaller than the decline in capital investment documented by Jens (2017) in the U.S. sample using quarterly data, and Julio and Yook (2012) in an international sample of firms. The differences in sign suggest that R&D expenditures do behave differently than capital expenditures when political uncertainty is high. Our results are consistent with Stein and Stone (2013), who find that R&D expenditures are positively related to general economic uncertainty.

We also examine the responsiveness of R&D to electoral uncertainty using Compustat quarterly data during close-election periods. Close-election periods are defined in the same way as in the annual data set. The downside to using quarterly data is that we are restricted to using a shorter sample time period as the quarterly data only go back to 1988, compared to 1976 for the annual Compustat data. However, the higher frequency allows for a closer examination of investment decisions around the actual election date. It also allows us to compare our results more directly with the previous literature using quarterly data such as Jens (2017).

We estimate investment regressions using the quarterly data based on a quarter-frequency version of the regression specification:

$$\begin{aligned} \text{R\&D Intensity}_{ijt} = & \alpha_i + \gamma_t + \beta_k \times \sum_{k=1}^4 \text{Close Election Quarter}_{kt} \\ & + \sum \varphi_i \mathbf{X}_{it} + \epsilon_{ijt}, \end{aligned} \quad (2)$$

where Close Election Quarter_{kt} is equal to one during an election year in quarters one through four, in close election cycles. We include firm and quarter fixed effects. We also control for Tobin's q, cash flow, and firm size in the main regressions. As with the annual regressions, we focus on close-election cycles. We also include capital expenditures to assets as a separate explanatory variable for comparison.

Table 3 reports the results of the baseline R&D and capital expenditure regressions using the quarterly Compustat data. Column 1 reports the R&D regression on the quarterly close election indicators and control variables with firm quarter fixed effects included. Column 2 reports the results with firm-level control variables included. Columns 3 and 4 report the same specifications but with capital expenditures as the dependent variable.

As we can see from Table 3, and consistent with the results using annual data, R&D investment is significantly higher than normal in quarters 2, 3, and 4 of

¹⁵ This is measured as the average effect relative to the mean (=0.0023/0.0876*100%) and median (=0.0023/0.0356*100%) R&D intensity.

Table 3
Differences between R&D and fixed investment: Quarterly data

	(1)	(2)	(3)	(4)
Dependent variable	R&D intensity	R&D intensity	Capex/Assets	Capex/Assets
Close election quarter 1	0.0001 (0.0003)	0.0001 (0.0003)	−0.0001 (0.0002)	−0.0001 (0.0002)
Close election quarter 2	0.0014*** (0.0003)	0.0013*** (0.0003)	−0.0006*** (0.0002)	−0.0005** (0.0002)
Close election quarter 3	0.0012*** (0.0003)	0.0011*** (0.0003)	−0.0008*** (0.0002)	−0.0007*** (0.0002)
Close election quarter 4	0.0012*** (0.0003)	0.0012*** (0.0003)	−0.0003 (0.0002)	−0.0002 (0.0002)
N	112,333	112,333	112,333	112,333
R ²	.2075	.2498	.1848	.1938
Firm controls	No	Yes	No	Yes
Firm FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes

This table reports investment regressions for both R&D and capital expenditures using Compustat quarterly data. The sample is over close election periods (margin of victory is less than 5%). Close Election election quarter dummies for quarters 1 through 4 are set equal to one during the specified quarter in an election year if the election is classified as “close.” Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. The variables of interest are the close election quarter indicators. We control for firm and quarter fixed effects and in some specifications include Tobin’s q, cash flow, and firm size as additional controls. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm and quarter levels. Data are for the period 1988 to 2021. * $p < .1$; ** $p < .05$; *** $p < .01$.

a close election year, by an average of 3.5% over the unconditional quarterly R&D mean. In contrast, capital expenditures are lower during the close election period, similar to the result in [Jens \(2017\)](#). Consistent with the view that the response to uncertainty shocks depends on the type of investment, we see that R&D increases when firms experience a political uncertainty shock, while capital expenditures tends to decline just before a highly contested election.

2.2 Degree of electoral uncertainty and R&D investment

The results reported in Tables 2 and 3 show a close election effect in firm-level R&D expenditures. In this section, we further analyze if close elections are different from nonclose elections, where the political uncertainty is smaller and firms have already incorporate expectations of political change. If uncertainty is driving the results, we expect the effects to be larger for close elections. Nonclose election cycles are characterized as 4-year periods, where the election year margin of victory is greater than 5%.

In addition to the margin of victory close-election measure that we use in our baseline results in the previous section, in this section we consider another election-based measure. It classifies elections as close (more uncertain) if the incumbent governors are not reelected. This could happen because they are not running, because of term-limit rules, or because of other exogenous factors such as scandals or other personal reasons. Previous studies extensively document that incumbency is an important predictor of election outcomes ([Ansola-behere and Snyder 2002](#)). Thus, it is reasonable to assume that if an incumbent governor is not reelected, the electoral uncertainty and competition

surrounding the election are likely to be high. Of the 598 elections in the sample, in 317 (approximately 53%) the incumbent governor is not reelected, which indicates a higher degree of political uncertainty surrounding the elections.

In columns 1 and 2, we perform subsample analysis by splitting the full sample into two groups according to whether the election cycle is classified as close or nonclose based on the same 5% margin of victory measure as in our baseline results. While the positive relationship between political uncertainty and R&D intensity is present in both close and nonclose election cycles, the coefficient is strongly statistically significant only for the close election cycles, and insignificant for nonclose election cycles. The coefficient estimate is about 3.2 times bigger in close than nonclose elections. This result suggests that close elections are more likely to provide significant uncertainty shocks as there is no average effect for elections that are not close.

In columns 3 and 4, we use the reelection status of the incumbent governor to measure election closeness. We again split the sample into two subsamples based on that status. We find that political uncertainty before the elections has a significant impact on R&D intensity only for firms in states where the incumbent is not reelected, that is, where the uncertainty is presumably higher.

One drawback of our close-election measure is that it is an *ex post* measure of highly uncertain elections. Ideally we would employ polling information to construct *ex ante* measures of election closeness. To that end, we collect recently available polling data, starting with the 2006 gubernatorial elections. We aggregate data from several polling agencies, which make predictions on the election outcome by stating their confidence that a given candidate and party will win the election. Their degree of uncertainty is measured by whether they consider a candidate and an election as “Safe,” “Solid,” “Likely,” “Lean,” “Tilt,” or a “Tossup.” We give each possible outcome a numerical value from 5 (“Safe”) to 0 (“Tossup”). A higher value of the polling variable in this case will indicate that it is closer to “Safe” and therefore has less political uncertainty. A lower value on the other hand will indicate that the variable is closer to a “Tossup” and therefore has greater political uncertainty.

In Table 4, panel B, we use the mean value of the predictions of all polling agencies in a given year in columns 1 and 3 and the median value of the predictions of the polling agencies in columns 2 and 4. A higher value of the consensus variable indicates greater political certainty (lower uncertainty). As dependent variables, we use two *ex post* measures related to the margin of victory. In columns 1 and 2, we use the raw measure of the margin of victory, which is the difference in the percentage of votes between the winner and the runner-up. Higher values of that variable indicate greater political certainty (lower uncertainty). In columns 3 and 4 we use the indicator variable equal to one if the margin of victory is less or equal to 5%, and zero otherwise. A value of one indicates greater political uncertainty.

Table 4
R&D in close versus nonclose election periods

A. R&D regressions				
	Dependent variable: R&D intensity			
	Close elect (1)	Nonclose elect (2)	Gov not reelected (3)	Gov reelected (4)
Election dummy	0.0023*** (0.0009)	0.0007 (0.0005)	0.0017** (0.0008)	−0.0002 (0.0010)
N	25,177	70,200	48,872	46,014
R ²	.8447	.8267	.8362	.8331
Firm controls	Yes	Yes	Yes	Yes
State controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
B. Ex ante and ex post measures of election closeness				
	Margin of victory	Margin of victory	Close election indicator	Close election indicator
	(1)	(2)	(3)	(4)
Mean polling consensus	5.1015*** (0.5071)		−0.1517*** (0.0220)	
Median polling consensus		4.3627*** (0.4657)		−0.1371*** (0.0204)
N	673	673	673	673
R ²	.7557	.7431	.5219	.5300

Panel A of this table examines whether the degree of electoral uncertainty amplifies the effect of political uncertainty on firm-level R&D intensity. In columns 1 and 2, we use election closeness to proxy for the degree of electoral uncertainty. Specifically, an election is classified as close if the margin of victory, defined as the vote difference between the first and second place candidates, is less than 5%. In columns 3 and 4, we use an indicator for whether an incumbent governor is reelected to proxy for electoral uncertainty. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include firm and year fixed effects. Standard errors are clustered at the firm level and corrected for heteroscedasticity. Standard errors are reported in square parentheses. To save space, we suppress the estimates of firm-specific and state economics control variables. * $p < .1$; ** $p < .05$; *** $p < .01$.

In columns 1 and 2 we document a positive relation between the *ex ante* polling measure of political certainty (consensus) and the margin of victory between the winner and runner up. In columns 3 and 4 we document a negative relation between the *ex ante* polling measure of political certainty (consensus) and the close election indicator. Overall, the results in Table 4, panel B, indicate a strong and positive relation between the *ex ante* and *ex post* measures of political uncertainty. These results provide validation of our choice of *ex post* measures of election closeness.

We interpret the findings in this section as strong evidence that the increase in R&D activity prior to an election is due to political uncertainty surrounding close elections. The effects are larger when election outcomes are more uncertain.

2.3 Politically sensitive industries during close-election periods

We turn now to studying additional cross-sectional evidence to further support our conjecture of a significant relationship between political uncertainty and

R&D investment. Our first step is to show that firms are likely to differ from each other with respect to their sensitivity to political uncertainty shocks. For example, both [Julio and Yook \(2012\)](#) and [Jens \(2017\)](#) find that the adverse effect of political uncertainty on firm investment is stronger in politically sensitive industries. [Boutchkova et al. \(2012\)](#) document that equity return volatility is higher around elections for politically sensitive industries. We expect that the election year effect in firm R&D is more pronounced for firms operating in politically sensitive industries, because these firms are more likely to face regulatory changes that affect their business operations and corporate decisions. Following the classification of industries by [Herron et al. \(1999\)](#), we classify firms operating in tobacco products, pharmaceuticals, health care services, defense, petroleum and natural gas, telecommunications, and transportation industries as politically sensitive, where industries are classified using the 48 industry classification scheme of [Fama and French \(1997\)](#). We then set a *PSI* dummy equal to one if a firm belongs to one of these politically sensitive industries, and zero otherwise.

To test the hypothesis, we perform subsample analyses for firms operating in politically sensitive industries and those in nonsensitive industries separately. We also interact the *PSI* dummy with the close election year dummy in the baseline R&D regression to directly compare the two types of industries. Table 5 reports the estimation results from this analysis. The coefficient estimate on the election dummy in column 1 of Table 2 translates into an economically significant 4.12% increase in politically sensitive firms' R&D-to-assets ratio in election years, relative to the mean R&D intensity for politically sensitive firms in nonelection years during close-election periods.¹⁶

Consistent with the view that politically sensitive industries are more exposed to political uncertainty, coefficient estimates in columns 1 and 2 indicate that politically sensitive industries experience approximately nine times higher increase in R&D intensity in an election year in close election cycles compared to politically nonsensitive industries, and the coefficient is statistically significant only for politically sensitive industries. The difference is statistically and economically significant. The interaction term reported in column 3 confirms that the difference is also statistically and economically significant.

2.4 Growth opportunities and technical uncertainty

R&D expenses represent all costs incurred during the financial year that relate to the development of new products or services. Not all R&D expenditures are the same. The models predicting a positive relationship between uncertainty and R&D rely on specific properties of the expenses. R&D expenditures that

¹⁶ This is measured as the average effect relative to the mean ($=0.0090/0.215 \times 100\%$) R&D intensity.

Table 5
Politically sensitive industries

	Dependent variable: R&D intensity		
	Politically sensitive (1)	Nonsensitive (2)	Interacted (3)
Close election dummy	.0090** (.0042)	.0010 (.0007)	.0007 (.0008)
Close election × PSIs			.0099** (.0039)
N	3,654	21,523	25,177
R ²	.8376	.8443	.8448
Firm controls	Yes	Yes	Yes
State controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

This table examines the cross-sectional variations of politically sensitive industries on firm-level R&D intensity in election years. Specifically, *politically sensitive industries (PSIs)* is an indicator variable set equal to one if firms fall into the following industries: Tobacco Products (5), Pharmaceuticals (13), Health Care Services (11), Defense (26), Petroleum and Natural Gas (30), Telecommunications (32) and Transportation (40), as used in [Herron et al. \(1999\)](#) and [Julio and Yook \(2012\)](#). Industries are classified using the 48 industry classification scheme of [Fama and French \(1997\)](#). In the first two columns, we perform subsample analysis by splitting the full sample into two subgroups according to the politically sensitive industry indicator and then examine the interactive effects between the election year dummy and the politically sensitive industry indicator in the last column. Each column is based on such a subsample, indicated by the column heading. The unit of observation is at firm-year level. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include the same control variables as in the baseline model in [Table 2](#) but the coefficients are omitted from the table to save space. [Table A1](#) in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. The regressions include firm and year fixed effects. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. To save space, we suppress the estimates of firm-specific and state economics control variables. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

acquire options on future investments, for example, should vary positively with uncertainty. Other properties include high technical uncertainty¹⁷ ([Pindyck 1993](#)) and investment lags or multistage investment ([Bar-Ilan and Strange 1996](#)).

This section empirically examines cross-sectional variation of growth opportunities and technical uncertainty in the relationship between firm R&D investment and political uncertainty shocks. We consider firms in high-tech industries as those having high growth opportunities, because their current market value heavily depends on future growth options. Therefore, their R&D investments are likely to be the type that generate these growth options. We also posit, based on previous research, that R&D projects of high tech firms evolve in multiple stages are also more likely to increase R&D when uncertainty is higher. [Bar-Ilan and Strange \(1996\)](#) show that investment lags can create an incentive to invest early when uncertainty is high as the opportunity cost of

¹⁷ In [Pindyck \(1993\)](#), technical uncertainty is resolved for firms active in R&D only, leading firms to invest sooner in an uncertain environment. A numerical experiment further leads [Pindyck \(1993\)](#) to conclude that “Thus for many investments, and particularly for large industrial projects where input costs fluctuate, increasing uncertainty is likely to depress investment. The opposite will be the case only for investments like R&D programs, where technical uncertainty is far more important.”

waiting now increases with uncertainty. [Bar-Ilan and Strange \(1998\)](#) show that when investment requires multiple stages, firms have an incentive to initiate exploratory investments, when uncertainty is high, to obtain an option to invest when the uncertainty resolves.

Based on the existing literature, we create two measures that proxy for high-tech firms. First, following [Brown, Fazzari, and Petersen \(2009\)](#), we create an indicator variable, “BFP High-tech,” equal to one for firms operating in seven high-tech industries: drugs, office equipment and computers, electronic components, communication equipment, scientific and medical instruments, and software, where industries are classified over the three-digit SIC codes. Second, we follow the work of [Hall, Jaffe, and Trajtenberg \(2005\)](#), who show that firms in the pharmaceutical, medical instruments, chemicals, computers, communications, and electrical industries have a larger percentage of their market value determined by R&D investments and patents. Following their research, we create an indicator variable, “HJT High-tech,” equal to one for firms operating in any of those industries, and zero otherwise.

Table 6 reports the regression results for the subsamples described above. Only high-tech firms show a positive and significant coefficient and on the close election indicator. The coefficient estimates on the interaction terms between the close election dummy and the proxy for high growth options are positive and statistically significant, suggesting that R&D expenditures for firms with more growth options have a higher sensitivity to political uncertainty. The results are economically significant. Using the BFP High-tech measure, we find that high-tech firms increase R&D intensity by an average of 3.00% in close election years relative to the average R&D intensity for the BFP High-tech subgroup. Using the HJT High-tech measure, we find that high-tech firms increase R&D intensity by an average of 4.48% in close election years relative to the average R&D intensity for the HJT High-tech subgroup. By contrast, this figure is close to zero, both economically and statistically insignificant, for low-tech firms, using both measures. The interaction terms in columns 3 and 6 are also economically and statistically significant, indicating that political uncertainty affects only high-tech firms.

2.5 Changes in R&D around close gubernatorial election cycles

In this section we examine R&D investment dynamics around the election year for the close election sample periods. We have seen that R&D is significantly higher during the year just before the election. We now examine whether the effect persists, goes away, or reverses after the electoral uncertainty is resolved. [Julio and Yook \(2012\)](#) and [Jens \(2017\)](#) find that capital expenditures decline prior to an election and then are significantly higher than normal in the year after the election. [Stokey \(2016\)](#) rationalizes this behavior by showing that when regulatory uncertainty is high, firms hold off on investments they would have accepted otherwise. These investments are tabled until the uncertainty is resolved. Investment after the uncertainty resolves is higher than

Table 6
Growth opportunities and high-growth firms

	Dependent variable: R&D intensity					
	BFP-high-tech			HJT-high-tech		
	High-tech (1)	Low-tech (2)	Interacted (3)	High-Tech (4)	Low-tech (5)	Interacted (6)
Close election dummy	0.0047*** (0.0017)	−0.0002 (0.0007)	0.0005 (0.0009) 0.0041*** (0.0014)	0.0066*** (0.0017)	−0.0004 (0.0009)	0.0008 (0.0009)
Election × BFP-high-tech indicator						
Election × HJT-high-tech indicator						0.0037** (0.0016)
N	11,643	13,534	25,177	10,063	15,114	25,177
R ²	.8171	.8315	.8447	.835	.8419	.8447
Constant	Yes	Yes	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

This table examines cross-sectional variation of growth opportunities of high-tech versus non-high-tech industries. We consider two classifications of high-tech firms. First, in columns 1–3, we follow [Brown, Fazzari, and Petersen \(2009\)](#) and create an indicator variable, *BFP-High-Tech*, set equal to one for firms operating in the seven high-tech industries, namely, drugs, office equipment and computers, electronic components, communication equipment, scientific and medical instruments, and software, where industries are classified over three-digit SIC code. Unlike other types of investment, R&D investment is highly concentrated in the seven high-tech industries. Second, in columns 4–6, we follow the work of [Hall, Jaffe, and Trajtenberg \(2005\)](#), we consider firms in the pharmaceuticals (13), medical instruments (12), chemicals (14), computers (35), communications (32), and electrical (22) industries that derive most of their value from R&D investments and patents. Industries are classified using the 48 industry classification scheme of [Fama and French \(1997\)](#). We therefore set a *HJT-high-tech* indicator to one if a firm belongs to one of these above-mentioned industries. The dependent variable in all regressions is R&D intensity, defined as the ratio of R&D expenditure to total assets. The regressions include the same control variables as in the baseline model in Table 2, but the coefficients are omitted from the table to save space. Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. We use the baseline regression specification and control for firm and year fixed effects. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

normal because the firm returns to its normal investment rate plus it invests in the projects that had been tabled, leading to a bust-and-boom dynamic for corporate investment. The dynamics of R&D, however, are not as clear. In this case, firms are increasing R&D and investing in projects they may have otherwise not considered. Given that R&D investment is theoretically considered either as a growth option or an option to abandon a project if the outcome is unfavorable, we do not predict that R&D expenditures will decline after the uncertainty is resolved.

We also examine changes in R&D in the years prior to the close election year to see if there are any preexisting trends that could put a dent in our causal story because it may indicate that R&D is changing even before the uncertainty increased. We explore the dynamics of R&D investment around the four-year election cycle [−1, 0, +1, +2] by estimating the following augmented baseline regression model that also include indicators for pre- and post-election years.

$$\begin{aligned}
\text{R\&D Intensity}_{ijt} = & \alpha_i + \gamma_t + \beta_0 \times \text{Preelection}_{jt}^{(-1)} + \beta_1 \times \text{Close Election}_{jt}^{(0)} + \beta_2 \\
& \times \text{Post-election}_{jt}^{(+1)} + \beta_3 \times \text{Post-election}_{jt}^{(+2)} \\
& + \sum \phi_i \mathbf{X}_{it} + \epsilon_{ijt},
\end{aligned} \tag{3}$$

where $\text{Close Election}_{jt}^{(0)}$ is the close election year dummy as defined in our baseline regression model in Equation (1). Similar to the election dummy, $\text{Preelection}_{jt}^{(-1)}$ is a dummy variable that equals one in the year right before a close election year. $\text{Post-election}_{jt}^{(+1)}$ and $\text{Post-election}_{jt}^{(+2)}$ are dummy variables that indicate the first year and second year following a close election year, respectively. These pre- and post-election dummies, along with the election dummy, are thus designed to capture the dynamics of R&D investment across the four-year election cycle. We include the control variables of the baseline regression specification (column 4, Table 2). As with all previous specifications, we also include firm and year fixed effects.

We report the estimation results in Table 7. In columns 1 and 2, we add the pre- and post-election dummy variables to the baseline regression model separately. In column 3, we include both dummy variables in addition to the close election year dummy. In column 4, we only include the two post-election dummies, in addition to the close election year dummy, in the baseline regression model. Across all specifications, we find that the election dummy variable remains positive and statistically significant, while the pre- and post-election dummies are insignificant. These results suggest that while firms increase R&D investment in close election years due to the increased political uncertainty created by upcoming gubernatorial elections, there is no discernible pattern in R&D in either the preelection years when political uncertainty is considered relatively weak or in the post-election years, when political uncertainty is resolved. In addition, the magnitude of R&D increase in close election years is similar to that reported in the baseline regression in Table 2. Unlike capital expenditures, R&D investments do not display any unusual pattern in the post-election period. This suggests that firms are investing in R&D ventures that would have not been accepted under normal periods of uncertainty.

2.6 Asset redeployability and substitution effects

In this section, we further examine the mechanism through which political uncertainty exerts a positive effect on R&D, but a negative influence on capital expenditures. Another possible channel that hasn't been adequately addressed in the literature is that of substitution. If a firm reduces one type of investment, it is left with extra cash that can be used to finance other types of investment. If it is harder to raise financing for R&D because of risk, for example, then firms

Table 7
Changes in R&D across the gubernatorial election cycle

	Dependent variable: R&D intensity			
	(1)	(2)	(3)	(4)
Preelection dummy (−1)	−0.0003 (0.0009)		0.0014 (0.0022)	
Close election dummy	0.0026** (0.0011)	0.0041*** (0.0011)	0.0054*** (0.0020)	0.0030*** (0.0010)
Post-election dummy (+1)		0.0040** (0.0018)	0.0045 (0.0029)	0.0030 (0.0021)
Post-election dummy (+2)				−0.00007 (0.0017)
N	23,442	23,267	21,645	21,327
R ²	.8533	.8436	.8519	.8426
Constant	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

This table investigate the dynamics of R&D investment across the four-year election cycle. In doing this, we follow [Julio and Yook \(2012\)](#) to estimate the following augmented baseline regression model that also include indicators for pre- and post-election years. The dependent variable in all specifications is R&D intensity, defined as the ratio of R&D expenditure to total assets. Variables of interests are the four election timing dummy variables. Close Election Dummy is the close election year dummy as defined in our baseline regression model. Similar to the election dummy, Preelection dummy is a dummy variable that equals one in the year right before an election year. Post-election dummy (+1) and Post-election dummy (+2) are dummy variables that indicate the first year and second year following a close election year, respectively. In columns 1 and 2, we add the pre- and post-election dummy variables to the baseline regression model separately. In column 3, we include both dummy variables in addition to the election year dummy. In column 4, we only include the two post-election dummies in the baseline regression model. We include firm and year fixed effects in all specifications. To save space, we suppress the estimates of firm-specific and state economics control variables. Table A1 in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

may use the cash saved from the reduced investment in fixed capital to finance R&D that otherwise would have gone unfunded. It could also be true that if heightened uncertainty makes R&D look relatively more attractive, then firms may reallocate funds away from capital expenditures and toward R&D. Thus, simply showing that capital expenditures decline on average when political uncertainty is high, while R&D expenditures increase during the same period of high uncertainty, is not enough to distinguish between irreversible investment effects and option value effects.

We examine the substitution hypothesis by exploring the interaction between asset redeployability and political uncertainty for both capital expenditures and R&D. [Kim and Kung \(2017\)](#) construct a measure of asset redeployability based on how many firms in the same industry use similar assets. Using the Bureau of Economic Analysis capital flow table, [Kim and Kung \(2017\)](#) construct their measure by calculating the proportion of firms that use a given asset. The measure varies across industries and over time. They show that firms with low asset redeployability (relatively more irreversible) reduce investment more when hit with an uncertainty shock.

Table 8
Asset redeployability and sensitivity to political uncertainty

Dependent variable	(1) R&D intensity	(2) Capex/Assets
Close election quarter 1	0.0006 (0.0004)	−0.0006 (0.0014)
Close election quarter 2	0.0012*** (0.0004)	0.0009 (0.0014)
Close election quarter 3	0.0013*** (0.0004)	−0.0057*** (0.0014)
Close election quarter 4	0.0012*** (0.0004)	−0.0020 (0.0015)
CEQ1 × Redeployability	0.0014 (0.0021)	0.0004 (0.0033)
CEQ2 × Redeployability	0.0009 (0.0020)	0.0018 (0.0032)
CEQ3 × Redeployability	0.0024 (0.0021)	0.0126*** (0.0032)
CEQ4 × Redeployability	0.0009 (0.0024)	0.0047 (0.0035)
Redeployability	−0.0304*** (0.0024)	−0.0230*** (0.0023)
N	94,256	94,256
R ²	.2680	.2016
Firm fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Firm controls	Yes	Yes

This table reports R&D and capital expenditure investment regressions using Compustat Quarterly data. The sample is over close election periods (margin of victory is less than 5%). *Close election* election-quarter (CEQ) dummies for quarters 1 through 4 are set equal to one during the specified quarter in an election year if the election is classified as “close.” The Redeployability variable is from [Kim and Kung \(2017\)](#). We include firm and year fixed effects in all specifications. Standard errors, reported in parentheses below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1988 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

We use the redeployability measure to test whether the positive effect of political uncertainty shocks on R&D expenditures is caused by the substitution mechanism described above. The idea here is that firms with low redeployability are those that are cutting back on capital expenditures and thus if the substitution effect dominates, then it should be exactly those firms that exhibit a higher R&D sensitivity to political uncertainty. We estimate both R&D and capital expenditure investment regressions and interact the redeployability measure with our election quarter indicators.

Table 8 reports the results of the regressions including the redeployability measure along with the interactions with our political uncertainty proxy. Column 1 reports the R&D regression. As before, the close election quarter indicators have a positive coefficient, showing that on average, R&D expenditures are higher in the quarters preceding a close election. The interaction terms with the [Kim and Kung \(2017\)](#) measure of asset redeployability are statistically insignificant, suggesting there is no difference in the R&D/uncertainty relationship based on the degree of asset redeployability. While it is difficult to completely rule out a role of substitution, our results suggest that it is unlikely to be the main mechanism. Column 2 reports the capital expenditure regression

to check whether the [Kim and Kung \(2017\)](#) result is present in our data. We find that in the third quarter of a close election year, firms with highly redeployable assets do not cut their investment as much as firms with a low degree of asset redeployability, consistent with [Kim and Kung \(2017\)](#).

Another result that suggests that the growth options hypothesis is the more likely mechanism compared to the substitution hypothesis can be seen in Table 3. Column 2 shows that, consistent with [Jens \(2017\)](#), capital expenditures decline significantly only in the third quarter of a close election year. Column 1, however, shows that the effect of political uncertainty has a longer duration on R&D expenditures. R&D is significantly higher in the three quarters surrounding the close election, suggesting that it is unlikely that the extra funds available during the third quarter are the source of the increase in R&D during the three-quarter period around the election.

2.7 Alternative measure of policy uncertainty

Our results up to this point have employed the timing of close gubernatorial elections as a measure of variation in political uncertainty over time and across states. The motivation for using elections as a proxy is that the timing of elections is fixed by law and in principle exogenous to economic conditions. The downside to using elections could be that they are relatively infrequent and may not lead to large changes in uncertainty. For robustness and external validity, in this section we employ the [Baker, Davis, and Levy \(2022\)](#) state-level Economic Policy Uncertainty Index (EPU) as an alternative measure of uncertainty. This index is constructed using textual analysis on newspapers in every state of the United States and Washington DC, including a total of about 3,500 newspapers. The index is based on newspaper coverage of policy-related uncertainty topics using a specified list of keywords. While there are concerns that text-based uncertainty indices are negatively correlated with the business cycle ([Baker, Bloom, and Davis 2016](#)), the state-level index provides a great deal of time-series variation across states. We follow the previous literature and use the natural logarithm of the EPU index as an alternative proxy for political uncertainty.

We first examine how the state-level EPU index varies across close election cycles. At the state level, we estimate a regression of the natural logarithm of the EPU index on the four close election quarter dummy variables and include state and year fixed effects. The results of the regression are reported in Panel A of Table 9. The state-level EPU index is significantly higher during quarters 3 and 4 of a close election cycle, with the highest increase in the index coming in the actual election quarter.

Panel B of Table 9 reports the results of both R&D and capital expenditure regressions on the state-level EPU index and control variables. Columns 1 and 2 report the baseline regressions for R&D and capital expenditures, respectively. Consistent with our previous results, we find that R&D expenditures have a positive relationship with the state-level EPU index, controlling for other

Table 9
Alternative proxy for policy uncertainty: State-level EPU index

A. <i>log(State-level EPU index)</i> around close elections					
	Q1	Q2	Q3	Q4	
Close election quarter coefficient	−0.0140 (0.0137)	0.0231 (0.0125)	0.0171*** (0.0055)	0.0653*** (0.0148)	
N	6,566				
R ²	0.8452				
B. <i>Investment regressions</i>					
Dependent variable	(1) R&D intensity	(2) Capex/Assets	(3) R&D intensity	(4) Capex/Assets	(5) R&D intensity
log(State-level EPU)	0.0006*** (0.0001)	−0.0043*** (0.0001)	0.0012*** (0.0003)	−0.0046*** (0.0008)	0.0016** (0.0007)
EPU × Redeployability			−0.0006 (0.0007)	0.0082*** (0.0020)	
Redeployability			−0.0012 (0.0035)	−0.0474*** (0.0085)	
Firm controls	Yes	Yes	Yes	Yes	Yes
N	450,714	450,714	362,500	362,500	112,333
R ²	.7182	.4116	.7165	.4374	.1590

This table reports R&D and capital expenditure investment regressions using Compustat Quarterly data with an alternative proxy for political uncertainty. Panel A reports state-level regressions of the state-level EPU index on close election quarters. The regressions include state and year fixed effects. Standard errors, reported in parentheses, are clustered by state. Panel B reports regression results of R&D intensity and capital expenditures. The variable of interest is the natural logarithm of the state-level EPU index of [Baker, Davis, and Levy \(2022\)](#). We include Tobin's q, cash flow, firm size, and firm and quarter fixed effects in all specifications. Column 5 reports the estimation results from a two-stage least squares model where close election indicators are used as instruments for the logarithm of the state-level EPU index. Table [A1](#) in the [appendix](#) defines the variable descriptions as well as the variable sources in detail. Standard errors, reported in square brackets below coefficient estimates, are clustered at the firm level and corrected for heteroscedasticity. Data are for the period 1988 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

determinants of R&D expenditures. We also find that capital expenditures are negatively correlated with the EPU index, consistent with [Gulen and Ion \(2016\)](#), who use the national level EPU index of [Baker, Bloom, and Davis \(2016\)](#) as their measure of policy uncertainty.

Columns 3 and 4 report R&D and capital expenditure regressions. In these columns, we include the redeployability measure of [Kim and Kung \(2017\)](#) as an additional explanatory variable and as an interaction with the EPU index. In column 3, we find that the effect of uncertainty on R&D does not have a significant interaction with redeployability, while column 4 shows that firms with higher asset redeployability reduce their investment less than other firms, consistent with the results of [Kim and Kung \(2017\)](#).

We note that the economic significance of the implied effect of the state-level EPU index during close election periods is lower than that of the close election periods alone as reported in [Table 3](#). The state-level EPU index increases by about 8.24% in quarters 3 and 4 of a close election cycle. The coefficient from column 1 from the R&D regression on state-level EPU implies a significant but small increase in R&D in response to an 8.24% increase in the index. This is likely because the EPU indices are

countercyclical (Baker, Bloom, and Davis 2016) and tend to be high during economic downturns. Ludvigson, Ma, and Ng (2021) show that high levels of the EPU index tend to lag downturns in economic activity, suggesting that R&D is responding negatively to economic downturns, but positively to high uncertainty, resulting in a net increase in R&D.

As an alternative, we also examine the relationship between R&D investment and the part of the state-level EPU index that changes during close elections. Using the same close election cycle sample employed in Table 3, we conduct a two-stage least squares regression where we obtain the predicted value of the natural logarithm of the state-level EPU index by regressing the index on the close election indicators in the first stage. In the second stage, we regress R&D investment on the predicted level of the log EPU index. Column 5 of Table 9 reports the estimation results. The coefficient on the log(state-level EPU) index is larger when using only the part of the index that is related to close elections. The coefficient is 0.0016, which is of a similar magnitude to that found in the baseline regressions using close election cycles as the proxy for political uncertainty.

2.8 Robustness tests

In this subsection, we perform a series of empirical tests to ensure that the preceding results are robust to various subsample and subperiod analyses, additional control variables, and alternative model specifications. We find that our results are still statistically and economically significant and similar to the baseline results.

First, in column 1, panel A of Table 10, we estimate the baseline R&D regression and conduct triple clustering by state, year and firm to control for multiple sources of within group correlation. The close election dummy remains positive and significant. Second, as mentioned earlier, over-the-counter (OTC) firms tend to be small technology stocks. The small size of these OTC firms may result in ratios that are highly variable and very large (in absolute value), which could disproportionately affect the results. To account for this problem, we perform subsample analysis by including only exchange traded firms (exchg = 11, 12, 14) and excluding OTC traded firms (exchg = 13, 19). Columns 2 of panel A report the results from this analysis. The coefficient estimates on the election dummy are significant and of larger magnitude as that in the overall sample. This finding indicates that our results are unlikely to be driven by the small OTC technology stocks.

Third, we perform several robustness tests on the sample selection. In columns 3 and 4 of panel A, we show that the positive effect of political uncertainty on R&D intensity remains significant, after removing observations during the dot-com period (1999–2000) or financial crisis period (2007–2008), respectively. Fourth, we ensure that our results are not driven by a small number of large states with a disproportionate representation of innovative firms in our sample. To that end, in column 5 we follow Lerner and Seru (2022) and

Table 10
Political uncertainty and R&D intensity: Robustness tests

A. Robustness tests for the baseline results in Table 2

	Dependent variable: R&D intensity				
	(1)	(2)	(3)	(4)	(5)
Election dummy	0.0023** (0.0011)	0.0025*** (0.0009)	0.0024** (0.0009)	0.0020*** (0.0009)	0.0026** (0.0007)
N	25,177	18,110	23,870	24,736	15,993
R ²	.8447	.8479	.8500	.8446	0.8531
Constant	Yes	Yes	Yes	Yes	Yes
Firm-specific controls	Yes	Yes	Yes	Yes	Yes
State economics controls	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

(Continued)

exclude firm-year observations from the three states with the highest number of innovative firms. These states are California, Massachusetts and New York, and they together comprise about 36% of the final sample observations. We then reestimate the models and find that our results are still strongly significant and not sensitive to the exclusion of firms from these states.

The political business cycle theory suggests that governors will increase expenditures when they are more uncertain of being reelected, and those higher expenditures, rather than the uncertainty itself will in turn drive the increase in R&D investment. We address these concerns in panel B of Table 10. Specifically, we study whether political uncertainty affects the total state expenditures (column 1), and the ratio of total state expenditures to general state revenues (column 2). We find that the gubernatorial election dummy variable is insignificant, suggesting that there is no significant change in state government expenditures during the election years. This result alleviates the concern that political business cycles might be driving the results

In unreported analyses, we conduct a number of additional robustness checks. First, we use R&D expenditures scaled by net sales as an alternative measure of R&D intensity and show that our main findings are robust to the new measure. Second, our results are robust to using an alternative measure of firm's headquarter state location based on the [García and Norli \(2012\)](#) data on the state-level operations of individual firms. [García and Norli \(2012\)](#) measure the state exposure of a firm's business operations to each U.S. state by conducting a textual analysis to record instances where state names occur in its annual 10-K filings. Since it is time varying, using this data set also allows us to control for firms that may be changing states over time. We show that our results are robust to this alternative definition of the relevant state.

Third, we repeat the analysis based on [Julio and Yook \(2012\)](#) international sample and also find that in their sample, R&D increases in election years. Fourth, we show that our conclusions remain unchanged if we only use a subsample of "innovative" firms that are required to have at least one patent

Table 10
(Continued)

B. Impact of electoral uncertainty on state government spending

	Total state expenditures (in millions USD)	Total expenditures to general revenues
	(1)	(2)
Election dummy	-0.7254 (1.6344)	0.0019 (0.0031)
Legislative election	-0.0936 (0.8910)	-0.0073** (0.0030)
Republican governor	-4.4872 (3.1123)	-0.0061 (0.0053)
Split legislature	-2.4263 (2.3874)	-0.0092 (0.0139)
GDP growth	0.3726 (0.2964)	-0.0046*** (0.0008)
Unemployment	1.4581 (1.0219)	0.0133*** (0.0029)
Constant	17.1335*** (5.4464)	1.0565*** (0.0163)
N	2,150	2,150
R ²	.405	.308
State FE	Yes	Yes
Year FE	Yes	Yes

This table presents robustness tests for the baseline R&D intensity regression results shown in Table 2. In panel A of Table 10, we use the baseline R&D regression specification and control for firm and year fixed effects across all columns. Standard errors, reported in parentheses below coefficient estimates, are clustered at the state level and corrected for heteroscedasticity in columns 2 to 5. In column 1, we reestimate the baseline regression model with standard errors triple clustered by state, firm, and year. Column 2 reports regression results based on the subsample of firms listed on NYSE, Amex, and Nasdaq only (Exchg = 11, 12, 14). In columns 3 and 4, we remove observations in the dot-com bubble period (i.e., 1999–2000) and financial crisis period (i.e., 2007–2008), respectively. Column 5 excludes firms headquartered in California, Massachusetts and New York to ensure that our results are not driven by a small number of large states with disproportionate representation of innovative firms in our sample. To save space, we suppress the estimates of firm-specific and state economics control variables. Data are for the period 1976 to 2020. * $p < .1$; ** $p < .05$; *** $p < .01$.

In panel B of Table 10, we examine the effects of electoral uncertainty induced by gubernatorial elections on state government spending. The dependent variable is total state expenditures in column 1 and total expenditures as a percentage of general revenues in column 2. Independent variables include additional dummy variables to indicate the timing of gubernatorial elections and state legislative elections, as well as the partisan status of state governors and legislatures. We also account for attributes of state economic conditions, including unemployment rate and gross domestic product (GDP) growth rate. Variable of interest is the election year indicator. In both columns, state and year fixed effects are included in the regression models, and standard errors, reported in parentheses below coefficient estimates, are clustered by state and year.

granted over the period from 1976 to 2010 in the U.S. patent data assembled by Kogan et al. (2017). Finally, we consider the possibility that mismeasurement in our proxy for Tobin’s q results in an inaccurate coefficient on the election dummy. To address this concern, we use the proxy-quality threshold test in Erickson and Whited (2005) to determine whether mismeasurement of Tobin’s q seriously affects our results. The results show that the potential measurement error in Tobin’s q is unlikely to bias the sign of our election dummy coefficients.

3. Conclusion

In this paper, we examine how political uncertainty affects R&D investment. Our purpose is not to add another corporate policy to the list of variables

adversely affected by policy uncertainty. Rather, we highlight that the theory of investment under uncertainty has different predictions for how investment responds to uncertainty, depending on the properties of the investment itself. Because many R&D projects represent the acquisition of a growth option and face high levels of technical uncertainty, the theory predicts that firms may increase R&D investments when political uncertainty is high. Investing in R&D gives a firm the option to invest if future conditions are favorable or abandon if they are not. Under uncertainty, the growth option has a large potential upside and a limited downside, leading to a positive relationship between R&D and uncertainty. Our results are consistent with these predictions.

In addition to the main result that R&D increases during close election years, we find that the effect is stronger when firms are more sensitive to political changes and when firms have more growth options and operate in industries with greater technological uncertainty. We also find that the R&D effect is not driven by a substitution away from capital expenditures when the firm experiences an uncertainty shock. All these results point to the growth options mechanism as the most likely explanation for our findings.

Our findings have important implications. First, we show that the relationship between investment and political uncertainty depends on the characteristics of the investment. Capital expenditures decrease, while R&D expenditures increase in the presence of political uncertainty. Second, it is not clear what the long-run effects of political uncertainty shocks are for the economy. As [Bar-Ilan and Strange \(1996\)](#) note, if investment is sometimes positively correlated with uncertainty, policymakers have less reason to be concerned with uncertainty. Warnings that policymakers should reduce lengthy policy debates and political uncertainty because of their negative effects should be made with caution and only after considering all types of investment.

Appendix Table A1. Variable descriptions

Variable	Definition	Source
R&D intensity variables		
R&D Intensity	Calculated as firms' research and development expenditure (XRD from Compustat) divided by book value of total assets (AT from Compustat), measured at the end of fiscal year t .	Compustat
Gubernatorial elections		
Post-election (−1)	Indicator variable takes on a value of one in the year right before a gubernatorial election year and zero in all other years.	CQE Library
Election (0)	Indicator variable takes on a value of one if a gubernatorial election occurred in that state in that year, and zero otherwise.	CQE Library
Post-election (+1)	Indicator variable takes on a value of one in the year right after a gubernatorial election year and zero in all other years.	CQE Library
Post-election (+2)	Indicator variable takes on a value of one in the second year after a gubernatorial election year and zero in all other years.	CQE Library
Close election	Indicator variable set equal to one if the victory margin, defined as the vote difference between the first and second place candidates, is less than 5%. We classify this type of elections as high uncertainty elections.	CQE Library
Close election cycle	Indicator variable set equal to one if a four-year election cycle [−1, 0, +1, +2] contains a close election in year 0, and zero otherwise	CQE Library
Incumbent reelected	Indicator variable set equal to one if the incumbent governors are reelected	CQE Library
Incumbent reelected	Indicator variable set equal to one if a four-year election cycle [−1, 0, +1, +2] contains a incumbent reelected election in year 0, and zero otherwise	CQE Library
Election cycle	The difference in the percentage of votes between the winner and the runner-up	CQE and Wikipedia
Margin of victory	Equal to the average prediction of the election outcome among polling agencies in a given election year.	Wikipedia
Mean polling consensus	Equal to the median prediction of the election outcome among polling agencies in a given election year.	Wikipedia
Firm-specific and state economics variables		
Tobin's q	Defined as [the market value of equity (PRCC_F × CSHO from Compustat) plus book value of assets (AT) minus book value of equity (CEQ from Compustat) minus balance sheet deferred taxes (TXDB from Compustat)] divided by book value of asset (AT), measured at the end of fiscal year t	Compustat
Cash flow	Firm's cash flows. It is defined as income before extraordinary items (IB from Compustat) plus depreciation and amortization (DP from Compustat) divided by book value of total assets (AT), measured at the end of fiscal year t	Compustat
ln(Age)	Defined as natural logarithm of one plus the number of years the firm is listed with a nonmissing stock price on Compustat	Compustat
ln(Sales)	Defined as natural logarithm of one plus the net sales/turnover (SALE from Compustat) measured at the end of fiscal year t	Compustat
Tangibility	Defined as total net property, plant and equipment (PPENT from Compustat) divided by book value of total asset (AT), measured at the end of fiscal year t	Compustat

(Continued)

(Continued)

Variable	Definition	Source
Profitability	Defined as earnings before interest, taxes, depreciation and amortization (EBITDA from Compustat) divided by book value of total asset (AT), measured at the end of fiscal year t	Compustat
Leverage	Firm's leverage ratio. It is defined as book value of debt (DLTT+DLC from Compustat) divided by book value of total assets (AT) measured at the end of fiscal year t	Compustat
Volatility	Monthly standard deviation of returns from daily CRSP stock returns	CRSP
Redeployability	Asset redeployability measure created by Kim and Kung (2017)	Kim and Kung 2017
GDP growth rate	State-level annual GDP growth rate, obtained from the U.S. Bureau of Economic analysis (BEA) website	BEA
Unemployment rate	State-level annual unemployment rate, obtained from the U.S. Bureau of Economic Analysis (BEA) website	BEA
HHI (or SIC HHI)	An industry-level measure of product market competition computed using firms' market shares as measured by three-digit SIC sales. Specifically, the SIC-based HHI is calculated as $HHI_{igt} = \sum_{i=1}^{N_g} S_{igt}^2$, where S_{igt} is the market share of firm i 's sales within a three-digit SIC industry g in year t and the summation is performed over the total number of firms N_g in that industry. When computing the HHI, we use all available Compustat firms, including those that do not report R&D. By construction, SIC HHI measures the degree of product market concentration and a lower product market concentration implies higher competition	Compustat
Politically sensitive industries (PSIs)	Indicator variable set equal to one for firms that belong to the following industries: Tobacco Products (5), Pharmaceuticals (13), Health Care Services (11), Defense (26), Petroleum and Natural Gas (30), Telecommunications (32) and Transportation (40), where industries are classified based on the 48 industry classification scheme of Fama and French (1997)	Herron et al. 1999
BFP-high-tech Dummy	Indicator variable set equal to one for firms operating in drugs, office equipment and computers, electronic components, communication equipment, scientific and medical instruments, and software	Brown, Fazzari, and Petersen 2009
HJT-High-Tech Dummy	Indicator variable set equal to one for firms operating in pharmaceuticals (13), medical instruments (12), chemicals (14), computers (35), communications (32), and electrical (22) industries. R&D programs in those industries are usually considered to feature a high degree of technical uncertainty and require long investment lags/multiple stages. Industries are classified using the 48 industry classification scheme of Fama and French (1997)	Hall, Jaffe, and Trajtenberg 2005
Governor party	Indicator variable that equals one if the governor is a Democrat in state j in year t , minus one if the governor is a Republican, and zero otherwise	CQE Library
Legislature Party	Indicator variable that equals one if Democrats control both chambers in state j in year t , minus one if Republicans control both chambers, and zero otherwise	CQE Library
Presidential election State EPU Index	Indicator variable that takes a value of one if a presidential election is held in that year State-level economic policy uncertainty (EPU) index created by Baker, Davis, and Levy (2022) Monthly index is averaged by calendar quarter	Hand-collected policyuncertainty.com

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